Stopping time and force

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain how a longer stop can reduce average force.

SUCCESS CRITERIA - I can...

1. Keep momentum change fixed in a fair comparison.
2. Compare force using impulse divided by time.
3. State why cushioning does not remove momentum change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “impulse” and “interval” to explain your model or result. Sentence starters are optional.

impulse: net force acting over time; change in momentum

interval: the time or range between two stated values

average: one value representing an interval or a set of values

01 Warm-up: bring back an earlier idea

A 3 kg cart moving right at 5 m/s stops in 2 s. Take right as positive.
Find signed impulse and average force.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

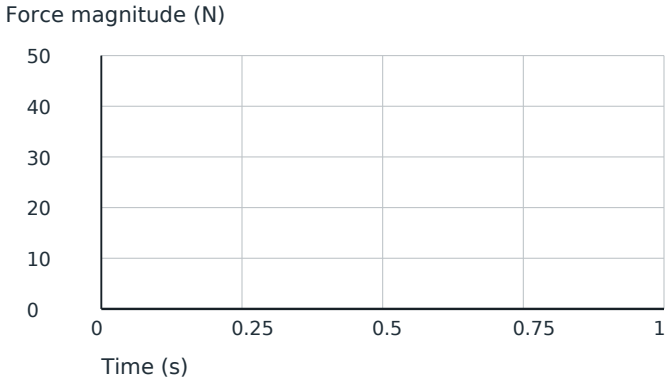
02 Worked example: explain the model

Reference: $|F_{avg}| = |\Delta p|/\Delta t$.
For the same momentum change, more stopping time lowers average force magnitude. Bars mean magnitude.

Worked reasoning: A stopping impulse has magnitude 8 N·s. Over 0.2 s, average force is 40 N; over 0.8 s, it is 10 N. Longer stopping time lowers the average force for the same momentum change.

Your turn: Draw both force-time rectangles with equal areas.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

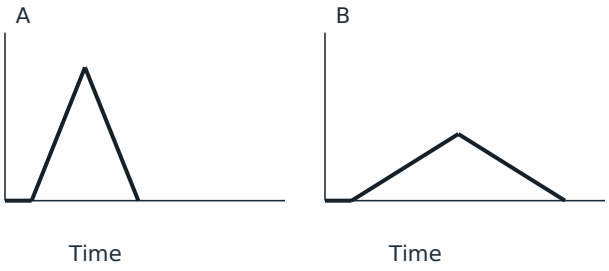
03 Read and interpret the model

The two ideal force pulses have equal area.
Identify the longer-duration, lower-average-force stop.

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice … . This supports … because … .

Force (same scales)



Marzano check - circle: ① ② ③

04 Apply the physics

A 0.060 kg ball traveling at 20 m/s comes to rest.
Compare two catches: 0.010 s and 0.050 s.
Find the impulse and average stopping-force magnitude for each.
Use the table to show what changes and what stays the same.

Word help: magnitude = size of a quantity without its direction

Sentence starter: Both … . They differ in … because … .

Stop	Time (s)	Impulse (N·s)	Average force (N)
A	____	____	____
B	____	____	____

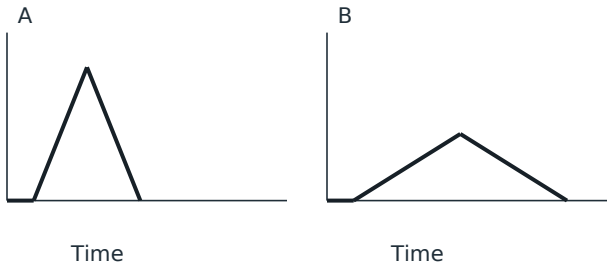
Marzano check - circle: ① ② ③

05 Explain your reasoning

A student says a cushion makes momentum disappear.
Explain where the momentum change comes from and what cushioning changes.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Force (same scales)



Same scales: compare impulse and peak force.

Marzano check - circle: ① ② ③

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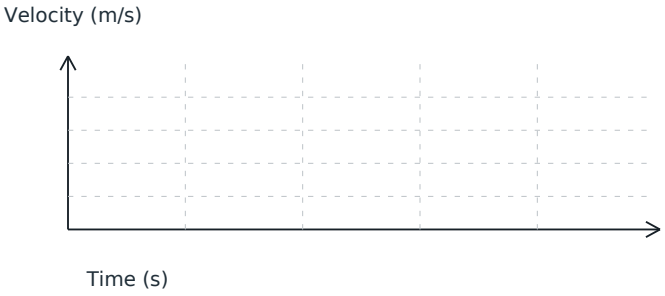
Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A downward-velocity fit goes from 0.4 m/s at 0.02 s to 3.34 m/s at 0.32 s.
Find g and explain why fitting an interval can be better than one noisy point.
Sketch the motion and label a positive direction.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Marzano check - circle: 1 2 3

07 Exit check

A 2 kg object slows from 6 to 2 m/s in 0.50 s.
Find impulse and average net force. If time becomes 1.0 s for the same velocity change, find the new average force.

Sentence starter: I found ... [unit]. This value means ... in this situation.

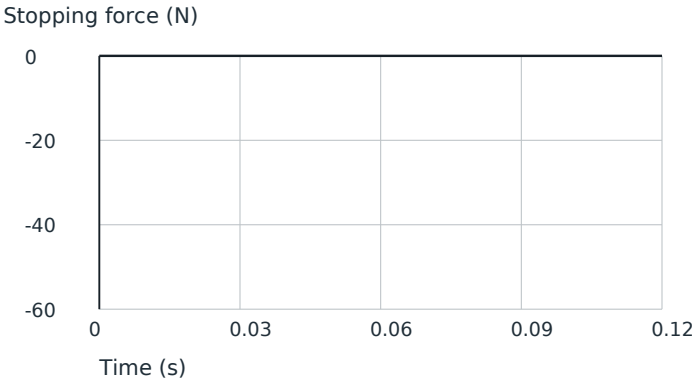
Trial	Duration	Average force
Original	Δt	F
New	____	____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 0.2 kg ball at 15 m/s stops in 0.03 s on one surface and 0.12 s on another.
Find average forces and design a triangular force-time model for the longer stop.

Sentence starter: My design shows ... . It meets the requirement by ... .



Marzano check - circle: 1 2 3